

User Hash Checker (UHC) User Guide

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Version 1

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1 Introduction

1.1 Scope and Purpose

This tool performs a weak-password audit by comparing existing user password hashes to a known wordlist (rockyou.txt). This helps identify potentially unsafe passwords that may violate organizational security standards.

The purpose of this user guide is to provide clear instructions for the use of the UHC tool, making the process as simple and straightforward as possible.

This guide covers the main workflow of the tool, such as:

- Selecting and loading a JSON file containing (hashed) user passwords.
- Loading a wordlist for password comparison.
- Running the password audit to identify weak passwords.
- Generating and interpreting the results report for human readability.

Assumptions and Prerequisites:

- This tool requires python to run. On some operating systems such as Kali Linux, python may already be installed. Users should have a compatible version of Python before running the tool.
- Users are to have their own JSON file full of users and pre-hashed passwords (as per below) + rockyou.txt downloaded and in the same folder as the UHC program.

```
▼ 0:
  user_name:      "George"
  user_password:  "2cf24dba5fb0a30e26e83b2ac5b9e29e1b161e5c1fa7425e73043362938b9824"
▼ 1:
  user_name:      "John"
  user_password:  "5994471abb01112afcc18159f6cc74b4f511b99806da59b3caf5a9c173cacfc5"
▼ 2:
  user_name:      "Sam"
  user_password:  "48a263f4af19b114032a57799c2b36fce09dbf1f8617ee7fb886be200275d89c"

[
  {
    "user_name": "George",
    "user_password": "2cf24dba5fb0a30e26e83b2ac5b9e29e1b161e5c1fa7425e73043362938b9824"
  },
  {
    "user_name": "John",
    "user_password": "5994471abb01112afcc18159f6cc74b4f511b99806da59b3caf5a9c173cacfc5"
  },
  {
    "user_name": "Sam",
    "user_password": "48a263f4af19b114032a57799c2b36fce09dbf1f8617ee7fb886be200275d89c"
  }
]
```

1.2 Process Overview/Workflow

UHC is designed to run a user selected JSON file containing usernames and SHA256 password hashes, load a wordlist, which in turn attempts to identify if a password matches to each of the stored hashes. The system guides the user through each step, whilst processing the data and outputting the results into a separate JSON file.

The software workflow follows this sequence:

1. Initial Setup:

- Core hashing functions, loading JSON files, and loading wordlist functions are setup.

2. Load Required Files:

- The user is prompted to enter the file path for a JSON file.
- Program attempts to load JSON file. If error occurs user is asked to try again.
- Program tries to find wordlist (rockyou.txt) in same folder as UHC program.
- Program attempts to load wordlist file. If error occurs user is told of error and ends program.

3. Password to Hash Matching:

- For each user record in the JSON file, the software will retrieve said username and password hash.
- The program hashes the entire wordlist once, stores the hashes in a dictionary, then checks each user's stored hash using a direct lookup.
- Each wordlist password entered is hashed using SHA256 to allow matching to occur.
- If a hash matches the stored hash, the password is marked as found.
- The result for each user (Match found/no match, and the password if applicable) is recorded.

4. Results Generated:

- The program output only shows whether a match was found or not and what user. The actual matching password (if found) is only shown in the results.json file, not in the console output.
- The results are then written to a new JSON file containing:
 - Username
 - Recovered Password (if applicable)
 - Indicator showing if a match was found or not

5. Error Handling:

- All file loading/writing functions include an error block for handling purposes.
- If the program encounters errors such as invalid file paths, unreadable files, and saving errors, user is told and sent back to retry.

6. Exiting the Program:

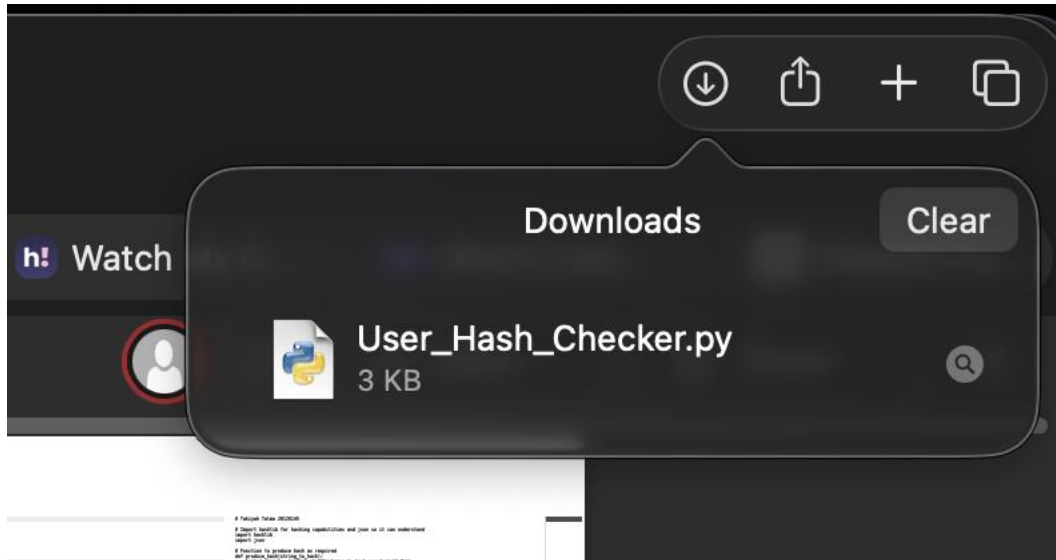
- The user may exit during the JSON prompt by entering [e].
- When exiting, the program will terminate after printing a goodbye message.

2 User Hash Checker Visual Workflow

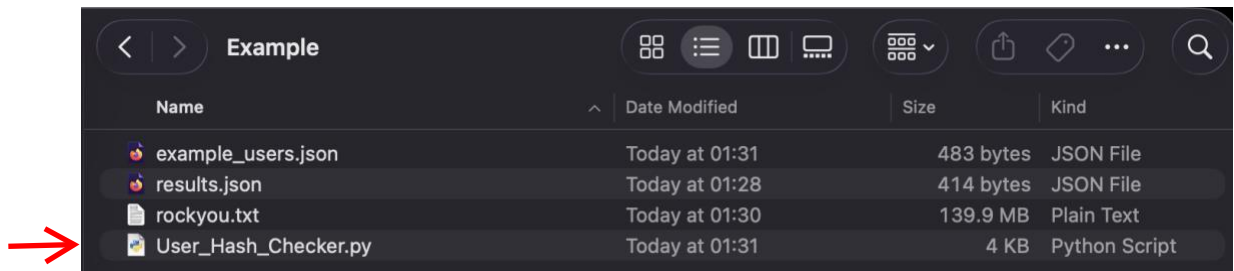
This visual workflow of UHC will show you through screenshots how to run this program.

2.1 Step-by-step procedure to run UHC

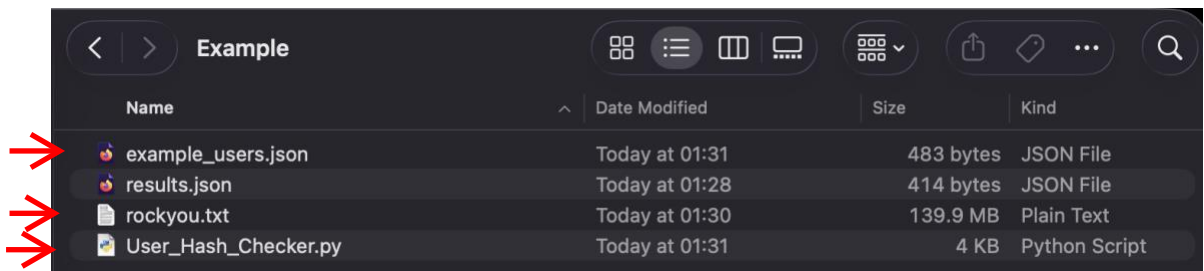
1. Download UHC program and find the file (most likely in your downloads folder) which can be easily located by going to your browser and clicking the downloads tab and then the folder/magnifying glass icon



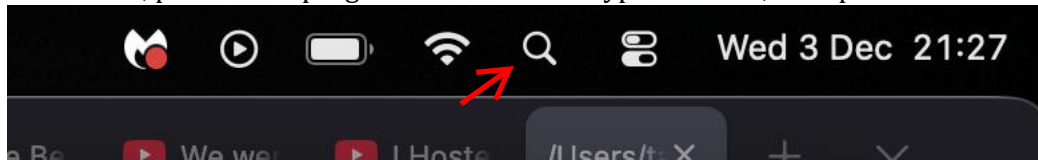
Browsers usually always have the download tab on the top right in a pop down menu as seen above



2. You will need to create a folder with the User_Hash_Checker.py file, your rockyou.txt file, and preferably your .json file



3. Run User_Hash_Checker_AS2.py through your command prompt/terminal
For macOS, press the top right search icon and type terminal, then press enter.



You will then be sent to an app like below, in which case you will type | `cd "/yourdirectoryhere/"` (yourdirectoryhere should look somewhat like what has been written below). You will then need to type | **python3 User_Hash_Checker.py**

```
[takiyah@Ts-Cy-MacBook ~ % cd "/Users/takiyah/Documents/Example"
[takiyah@Ts-Cy-MacBook Example % python3 User_Hash_Checker.py
Enter your json file path or [E] to exit: █
```

For **Windows** (Command Prompt) it is similar, you will either press your Windows key on your keyboard or press the search button on the bottom middle (bottom left if you are on windows 10 or has been configured as such), then typing command prompt and pressing enter (open)

You will then write | `cd C:\Users\YourUsername\Documents` | (Your directory inputted instead) and then | **python User_Hash_Checker.py**

cd (change directory) line is to get into your file on your terminal

python3 (filename) is to run your python file

4. Input your wanted JSON files pathway then press enter

```
Enter your json file path or [E] to exit: example_users.json
```

If the file is in the same directory as the UHC program, you can likely just run the "file name".json

```
Enter your json file path or [E] to exit: /Users/takiyah/Documents/Example/example_users.json
```

If the file is outside of the same directory, you will need to input the complete path

5. Then there will be no new typeable line for a short moment as the wordlist will be getting read through, wait during this time unless program is prolonged excessively
6. You may encounter an error (see below) if you have not correctly inputted the json file or other issues of that regard

```
Enter your json file path or [E] to exit: incorrect_example.json
Error loading JSON file, make sure it is written correctly
```

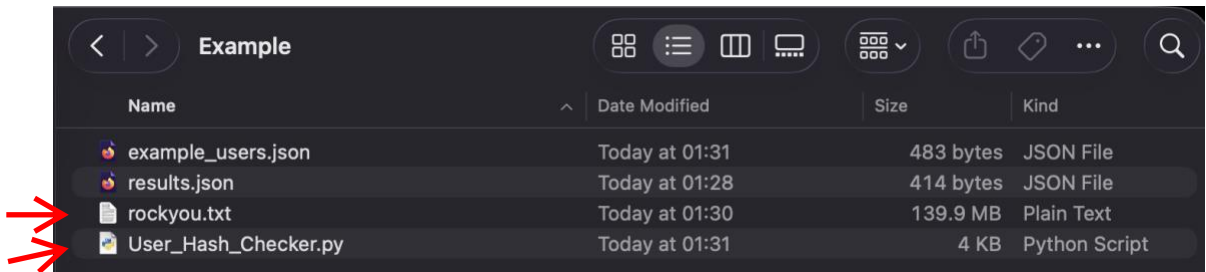
To deal with this or make it easier, please input json file into a folder with the program file and make sure the name is written correctly whilst adding ".json" (remove quotations and spaces) at the end

Example: UHC_Program.json

7. After inputting your .json file, you may be hit with another error stating "rockyou.txt has not been loaded" (see below)

```
Enter your json file path or [E] to exit: example_users.json
Rockyou.txt file not found, make sure it is in the same folder as UHC program
```

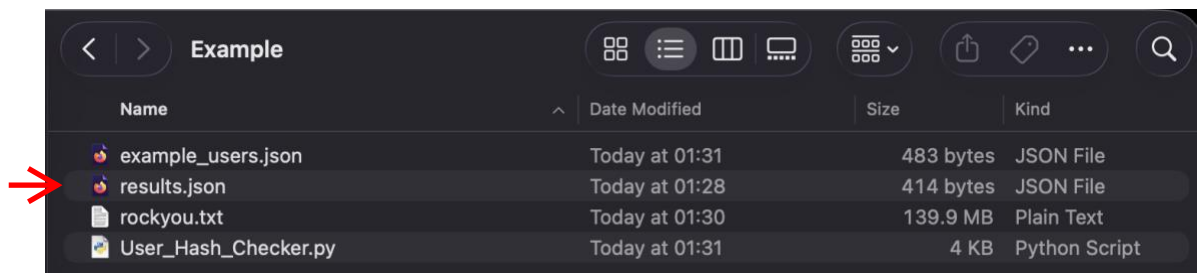
Because of how this program has been made, you will need the rockyou.txt file to be downloaded in the same folder as the UHC program



8. Once results have popped up, it will show you who has and has not been 'found' as well as telling you [results.json] has been saved, restarting the loop

```
Enter your json file path or [E] to exit: /Users/takiyah/Documents/Example/example_users.json
Alice: Match Found
Bob: Match Found
Bill: Match Found
Ted: Match Found
Results saved to results.json
```

9. You can then check the results.json file which can be found where your program has been saved



10. Opening the results.json file will prompt you with the users, passwords (if found), and if a match was found

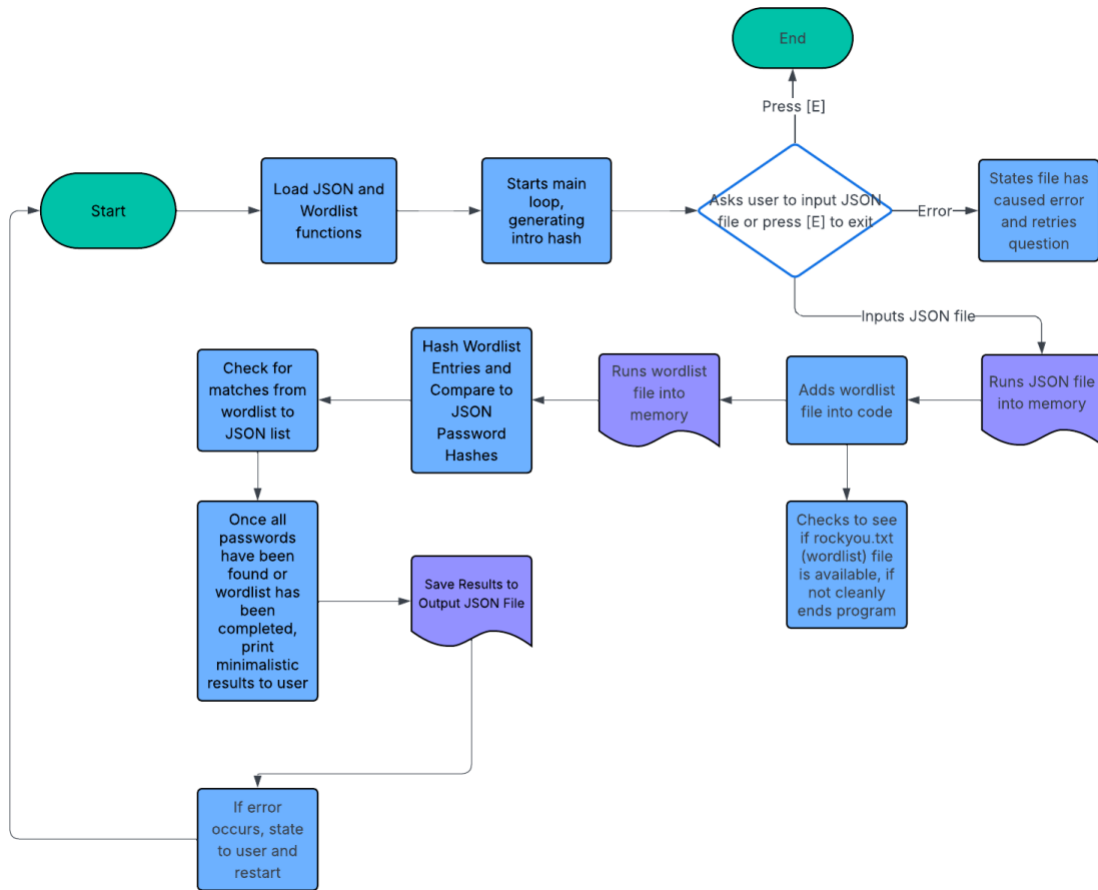
```
0:
  username: "Alice"
  password: "rainbow"
  password_found: true
1:
  username: "Bob"
  password: "batman"
  password_found: true
2:
  username: "Bill"
  password: "liverpool"
  password_found: true
3:
  username: "Ted"
  password: "thegame"
  password_found: true
```

```
▼ 0:
  username:      "George"
  password:      "hello"
  password_found: true
▼ 1:
  username:      "John"
  password:      "12345"
  password_found: true
▼ 2:
  username:      "Sam"
  password:      ""
  password_found: false
```

3 Index

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Table of Contents	Page numbers of important information
Scope and Purpose of UHC	Why this program and user guide has been made
Process overview and workflow of UHC	Text based workflow of UHC for understanding on how to use this program
Workflow Flowchart	Flowchart of UHC's code
Visual Workflow of UHC	Screenshots of how to run UHC

4 Appendix



[Flowchart of Workflow]